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Section: BCS-243D

Syllabus: Database Systems Lab

Syllabus Code: CL2005

Instructor: Sir Osama Tahir

Q1)

Query:

```
SELECT e.first_name, d.department_name
FROM HR.employees e
INNER JOIN HR.departments d ON e.department_id = d.department_id;
```

Snapshot:

Worksheet Query Builder

```sql
SELECT e.first_name, d.department_name
FROM HR.employees e
INNER JOIN HR.departments d ON e.department_id = d.department_id;
```

Query Result

SQL | Fetched 50 rows in 0.138 seconds

	FIRST_NAME	DEPARTMENT_NAME
1	Jennifer	Administration
2	Pat	Marketing
3	Michael	Marketing
4	Sigal	Purchasing
5	Karen	Purchasing
6	Shelli	Purchasing
7	Den	Purchasing
8	Alexander	Purchasing
9	Guy	Purchasing
10	Susan	Human Resources
11	Kevin	Shipping
12	Jean	Shipping
13	Adam	Shipping
14	Timothy	Shipping
15	Ki	Shipping
16	Girard	Shipping
17	Douglas	Shipping
18	Vance	Shipping
19	Payam	Shipping
20	Renske	Shipping
21	Britney	Shipping
22	Jennifer	Shipping
23	Julia	Shipping
24	Curtis	Shipping

Q2)

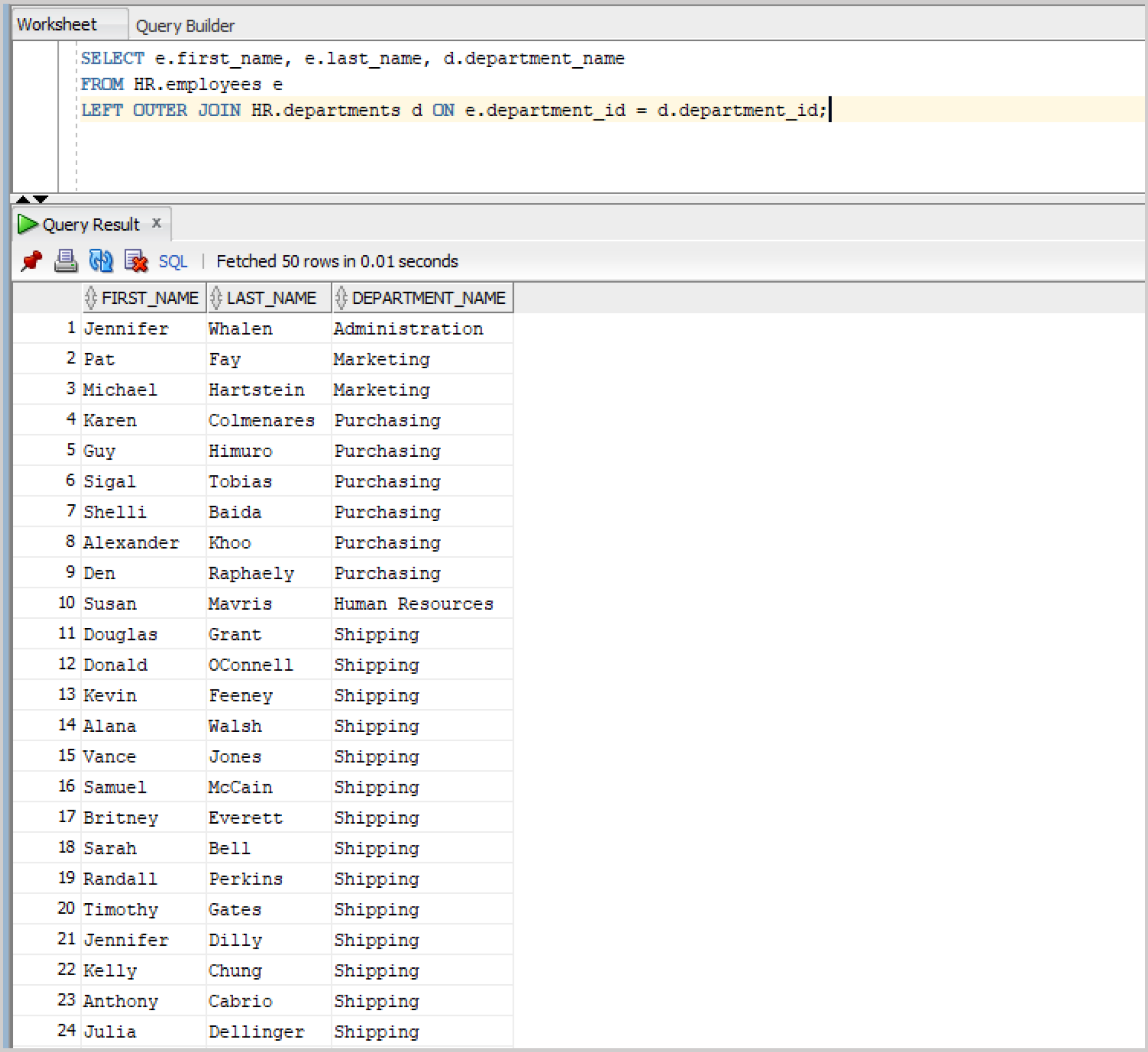
Query:

SELECT e.first_name, e.last_name, d.department_name

FROM HR.employees e

LEFT OUTER JOIN HR.departments d ON e.department_id = d.department_id;

Snapshot:



	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Jennifer	Whalen	Administration
2	Pat	Fay	Marketing
3	Michael	Hartstein	Marketing
4	Karen	Colmenares	Purchasing
5	Guy	Himuro	Purchasing
6	Sigal	Tobias	Purchasing
7	Shelli	Baida	Purchasing
8	Alexander	Khoo	Purchasing
9	Den	Raphaely	Purchasing
10	Susan	Mavris	Human Resources
11	Douglas	Grant	Shipping
12	Donald	OConnell	Shipping
13	Kevin	Feeney	Shipping
14	Alana	Walsh	Shipping
15	Vance	Jones	Shipping
16	Samuel	McCain	Shipping
17	Britney	Everett	Shipping
18	Sarah	Bell	Shipping
19	Randall	Perkins	Shipping
20	Timothy	Gates	Shipping
21	Jennifer	Dilly	Shipping
22	Kelly	Chung	Shipping
23	Anthony	Cabrio	Shipping
24	Julia	Dellinger	Shipping

Q3)

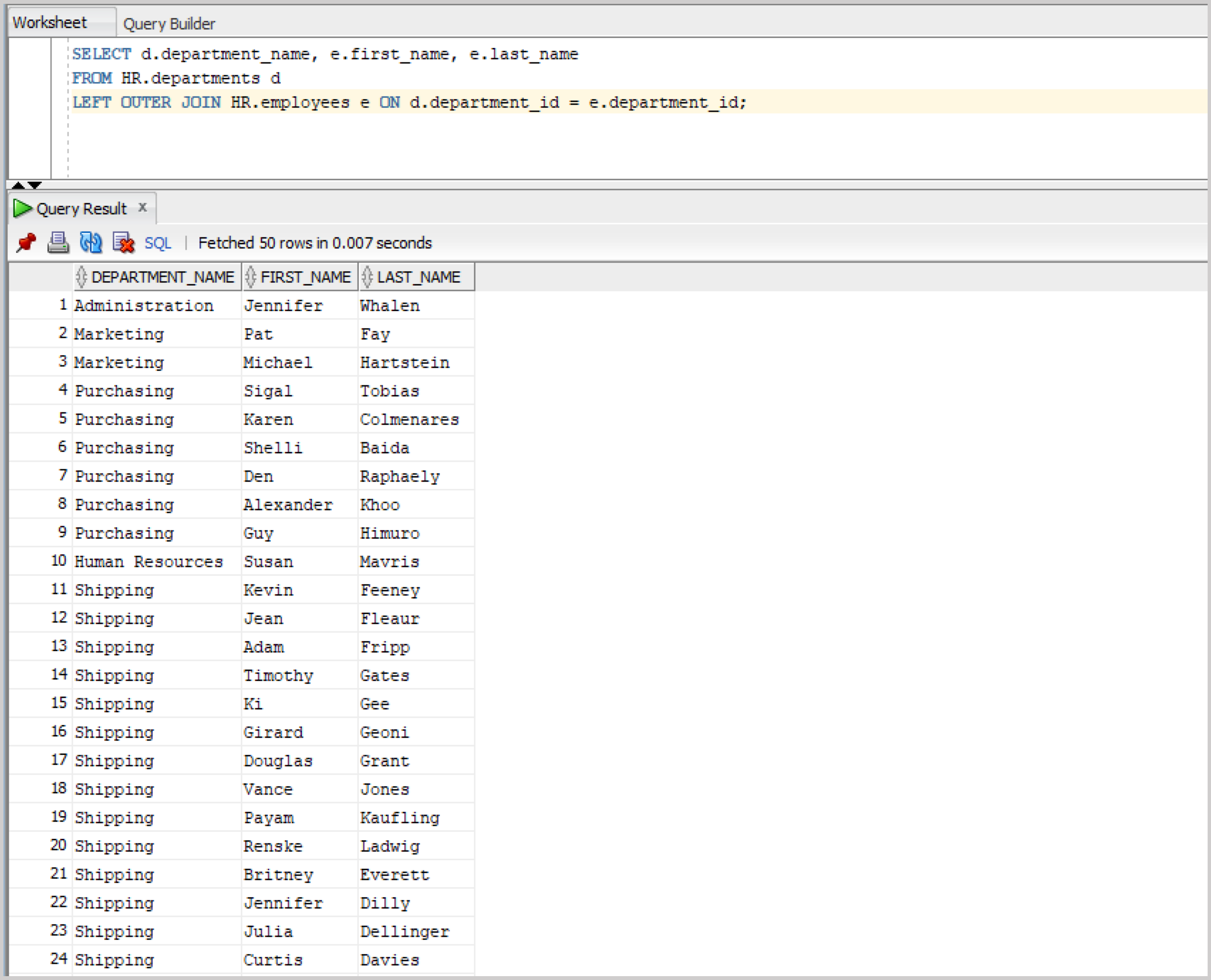
Query:

SELECT d.department_name, e.first_name, e.last_name

FROM HR.departments d

LEFT OUTER JOIN HR.employees e ON d.department_id = e.department_id;

Snapshot:



	DEPARTMENT_NAME	FIRST_NAME	LAST_NAME
1	Administration	Jennifer	Whalen
2	Marketing	Pat	Fay
3	Marketing	Michael	Hartstein
4	Purchasing	Sigal	Tobias
5	Purchasing	Karen	Colmenares
6	Purchasing	Shelli	Baida
7	Purchasing	Den	Raphaely
8	Purchasing	Alexander	Khoo
9	Purchasing	Guy	Himuro
10	Human Resources	Susan	Mavris
11	Shipping	Kevin	Feeney
12	Shipping	Jean	Fleaur
13	Shipping	Adam	Fripp
14	Shipping	Timothy	Gates
15	Shipping	Ki	Gee
16	Shipping	Girard	Geoni
17	Shipping	Douglas	Grant
18	Shipping	Vance	Jones
19	Shipping	Payam	Kaufling
20	Shipping	Renske	Ladwig
21	Shipping	Britney	Everett
22	Shipping	Jennifer	Dilly
23	Shipping	Julia	Dellinger
24	Shipping	Curtis	Davies

Q4)

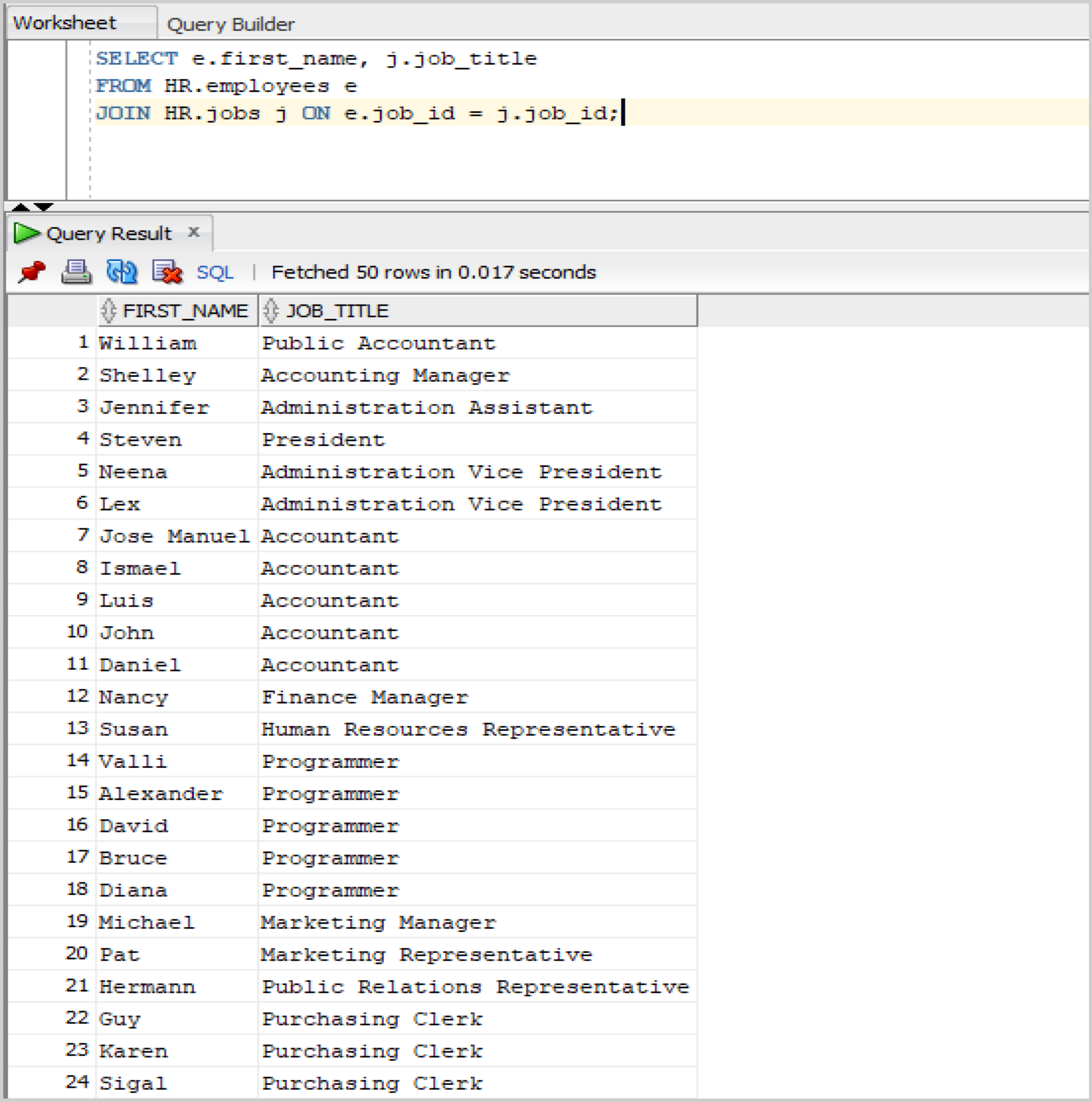
Query:

SELECT e.first_name, j.job_title

FROM HR.employees e

JOIN HR.jobs j ON e.job_id = j.job_id;

Snapshot:



Q5)

Query:

SELECT e.first_name || ' ' || e.last_name AS full_name, department_id, d.location_id

FROM HR.employees e

JOIN HR.departments d USING (department_id);

Snapshot:

```
SELECT e.first_name || ' ' || e.last_name AS full_name, department_id, d.location_id
FROM HR.employees e
JOIN HR.departments d USING (department_id);
```

Query Result

SQL | Fetched 50 rows in 0.012 seconds

	FULL_NAME	DEPARTMENT_ID	LOCATION_ID
1	Jennifer Whalen	10	1700
2	Pat Fay	20	1800
3	Michael Hartstein	20	1800
4	Sigal Tobias	30	1700
5	Karen Colmenares	30	1700
6	Shelli Baida	30	1700
7	Den Raphaely	30	1700
8	Alexander Khoo	30	1700
9	Guy Himuro	30	1700
10	Susan Mavris	40	2400
11	Kevin Feeney	50	1500
12	Jean Fleaur	50	1500
13	Adam Fripp	50	1500
14	Timothy Gates	50	1500
15	Ki Gee	50	1500
16	Girard Geoni	50	1500
17	Douglas Grant	50	1500
18	Vance Jones	50	1500
19	Payam Kaufling	50	1500
20	Renske Ladwig	50	1500
21	Britney Everett	50	1500
22	Jennifer Dilly	50	1500
23	Julia Dellinger	50	1500
24	Curtis Davies	50	1500

Q6)

Query:

SELECT first_name, last_name, department_name

FROM HR.employees

NATURAL JOIN HR.departments;

Snapshot:

Worksheet Query Builder

```
SELECT first_name, last_name, department_name
FROM HR.employees
NATURAL JOIN HR.departments;
```

Query Result

SQL | All Rows Fetched: 32 in 0.004 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Neena	Kochhar	Executive
2	Lex	De Haan	Executive
3	Bruce	Ernst	IT
4	David	Austin	IT
5	Valli	Pataballa	IT
6	Diana	Lorentz	IT
7	Daniel	Faviet	Finance
8	John	Chen	Finance
9	Ismael	Sciarra	Finance
10	Jose Manuel	Urman	Finance
11	Luis	Popp	Finance
12	Alexander	Khoo	Purchasing
13	Shelli	Baida	Purchasing
14	Sigal	Tobias	Purchasing
15	Guy	Himuro	Purchasing
16	Karen	Colmenares	Purchasing
17	Laura	Bissot	Shipping
18	Mozhe	Atkinson	Shipping
19	James	Marlow	Shipping
20	TJ	Olson	Shipping
21	Peter	Tucker	Sales
22	David	Bernstein	Sales
23	Peter	Hall	Sales
24	Christopher	Olsen	Sales

Q7)

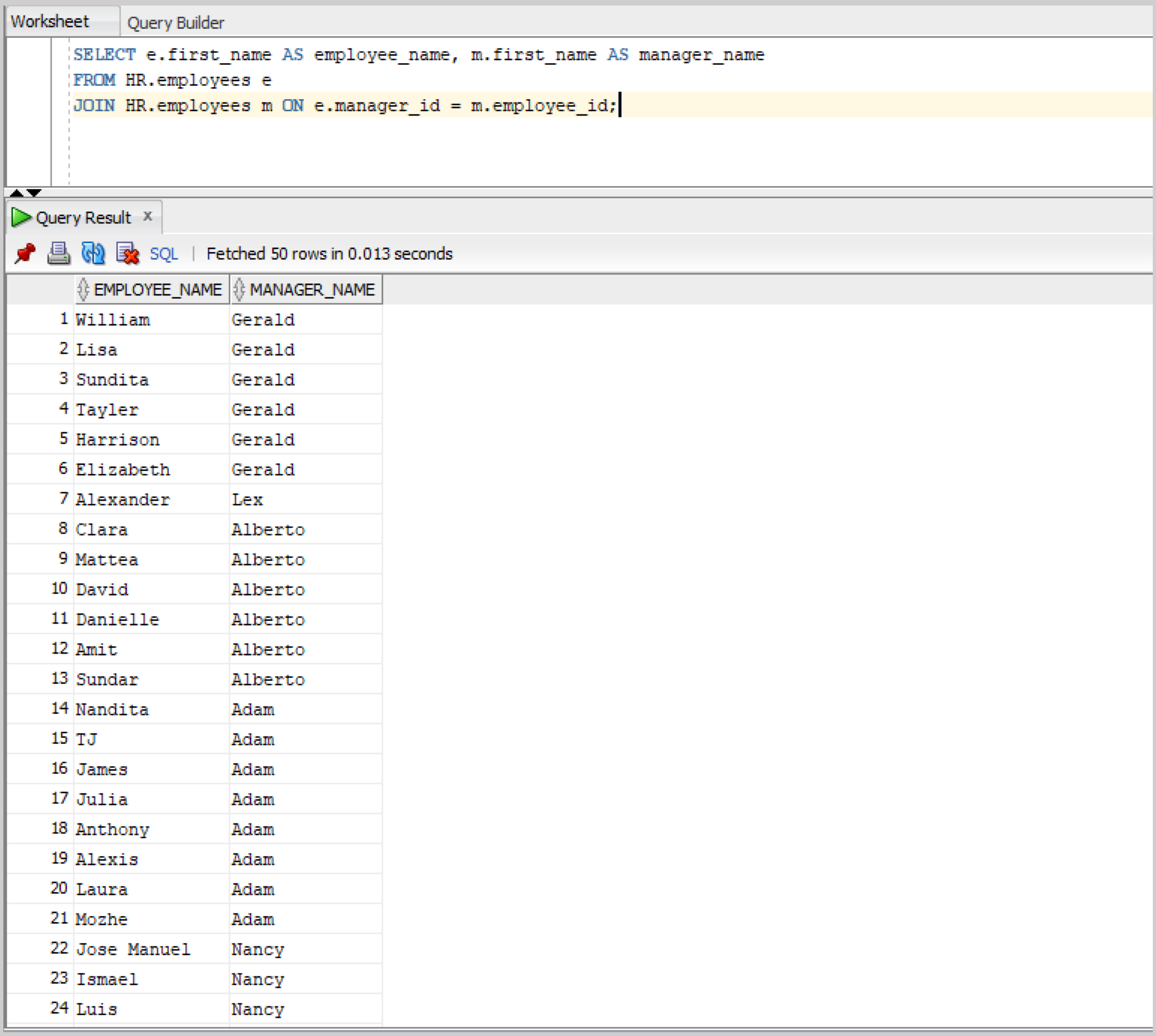
Query:

SELECT e.first_name AS employee_name, m.first_name AS manager_name

FROM HR.employees e

JOIN HR.employees m ON e.manager_id = m.employee_id;

Snapshot:



	EMPLOYEE_NAME	MANAGER_NAME
1	William	Gerald
2	Lisa	Gerald
3	Sundita	Gerald
4	Tayler	Gerald
5	Harrison	Gerald
6	Elizabeth	Gerald
7	Alexander	Lex
8	Clara	Alberto
9	Mattea	Alberto
10	David	Alberto
11	Danielle	Alberto
12	Amit	Alberto
13	Sundar	Alberto
14	Nandita	Adam
15	TJ	Adam
16	James	Adam
17	Julia	Adam
18	Anthony	Adam
19	Alexis	Adam
20	Laura	Adam
21	Mozhe	Adam
22	Jose Manuel	Nancy
23	Ismael	Nancy
24	Luis	Nancy

Q8)

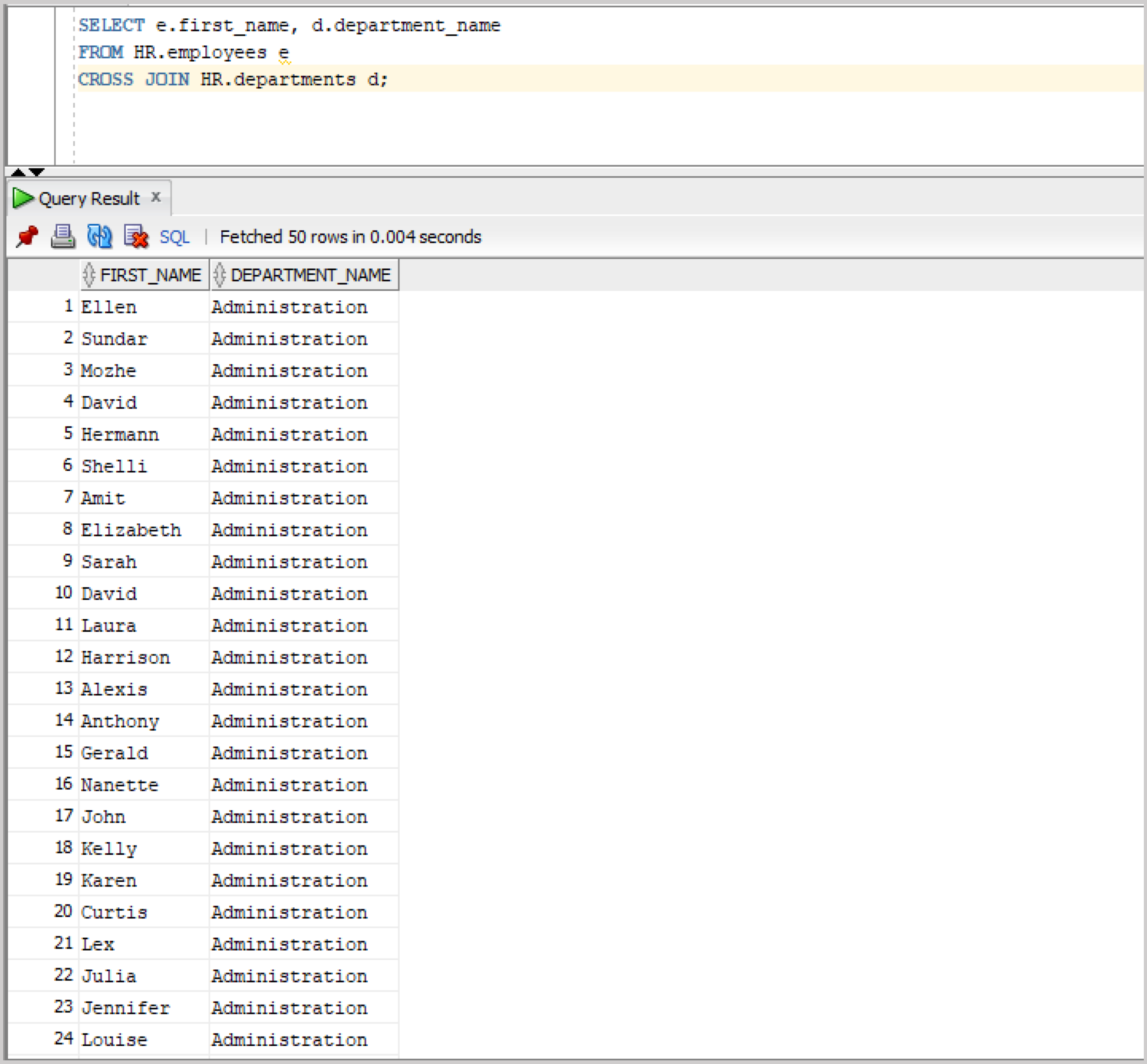
Query:

SELECT e.first_name, d.department_name

FROM HR.employees e

CROSS JOIN HR.departments d;

Snapshot:



```
SELECT e.first_name, d.department_name
FROM HR.employees e
CROSS JOIN HR.departments d;
```


	FIRST_NAME	DEPARTMENT_NAME
1	Ellen	Administration
2	Sundar	Administration
3	Mozhe	Administration
4	David	Administration
5	Hermann	Administration
6	Shelli	Administration
7	Amit	Administration
8	Elizabeth	Administration
9	Sarah	Administration
10	David	Administration
11	Laura	Administration
12	Harrison	Administration
13	Alexis	Administration
14	Anthony	Administration
15	Gerald	Administration
16	Nanette	Administration
17	John	Administration
18	Kelly	Administration
19	Karen	Administration
20	Curtis	Administration
21	Lex	Administration
22	Julia	Administration
23	Jennifer	Administration
24	Louise	Administration

Q9)

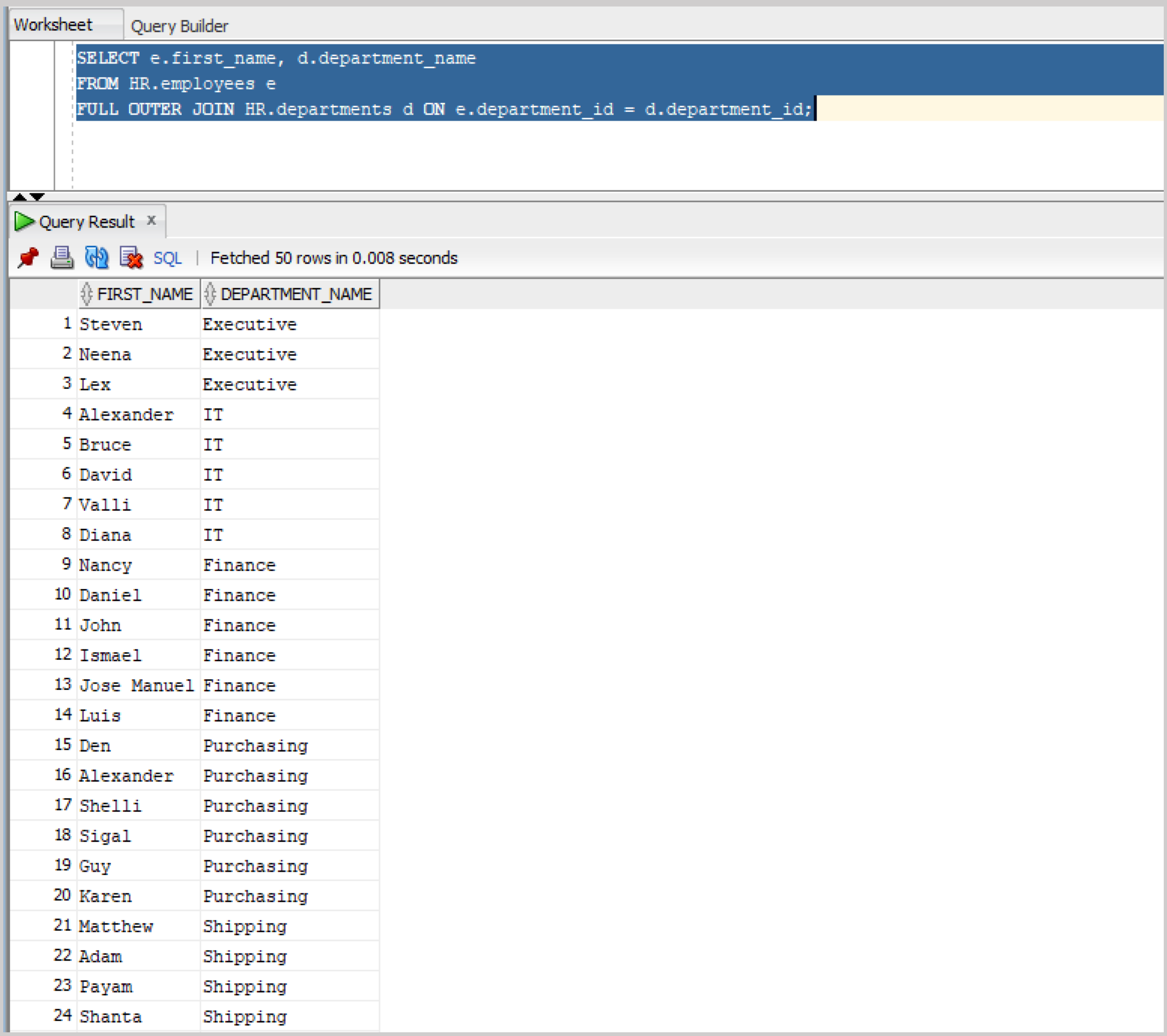
Query:

SELECT e.first_name, d.department_name

FROM HR.employees e

FULL OUTER JOIN HR.departments d ON e.department_id = d.department_id;

Snapshot:



	FIRST_NAME	DEPARTMENT_NAME
1	Steven	Executive
2	Neena	Executive
3	Lex	Executive
4	Alexander	IT
5	Bruce	IT
6	David	IT
7	Valli	IT
8	Diana	IT
9	Nancy	Finance
10	Daniel	Finance
11	John	Finance
12	Ismael	Finance
13	Jose Manuel	Finance
14	Luis	Finance
15	Den	Purchasing
16	Alexander	Purchasing
17	Shelli	Purchasing
18	Sigal	Purchasing
19	Guy	Purchasing
20	Karen	Purchasing
21	Matthew	Shipping
22	Adam	Shipping
23	Payam	Shipping
24	Shanta	Shipping

Q10)

Query:

SELECT e.first_name, d.department_name, j.job_title

FROM HR.employees e

JOIN HR.departments d ON e.department_id = d.department_id

JOIN HR.jobs j ON e.job_id = j.job_id;

Snapshot:

```
SELECT e.first_name, d.department_name, j.job_title
FROM HR.employees e
JOIN HR.departments d ON e.department_id = d.department_id
JOIN HR.jobs j ON e.job_id = j.job_id;
```

Query Result ×

SQL | Fetched 50 rows in 0.009 seconds

	FIRST_NAME	DEPARTMENT_NAME	JOB_TITLE
1	Jennifer	Administration	Administration Assistant
2	Pat	Marketing	Marketing Representative
3	Michael	Marketing	Marketing Manager
4	Den	Purchasing	Purchasing Manager
5	Karen	Purchasing	Purchasing Clerk
6	Guy	Purchasing	Purchasing Clerk
7	Sigal	Purchasing	Purchasing Clerk
8	Alexander	Purchasing	Purchasing Clerk
9	Shelli	Purchasing	Purchasing Clerk
10	Susan	Human Resources	Human Resources Representative
11	Matthew	Shipping	Stock Manager
12	Adam	Shipping	Stock Manager
13	Payam	Shipping	Stock Manager
14	Shanta	Shipping	Stock Manager
15	Kevin	Shipping	Stock Manager
16	John	Shipping	Stock Clerk
17	Stephen	Shipping	Stock Clerk
18	Renske	Shipping	Stock Clerk
19	Hazel	Shipping	Stock Clerk
20	Ki	Shipping	Stock Clerk
21	Michael	Shipping	Stock Clerk
22	Jason	Shipping	Stock Clerk
23	Peter	Shipping	Stock Clerk
24	James	Shipping	Stock Clerk

Q11)

Query:

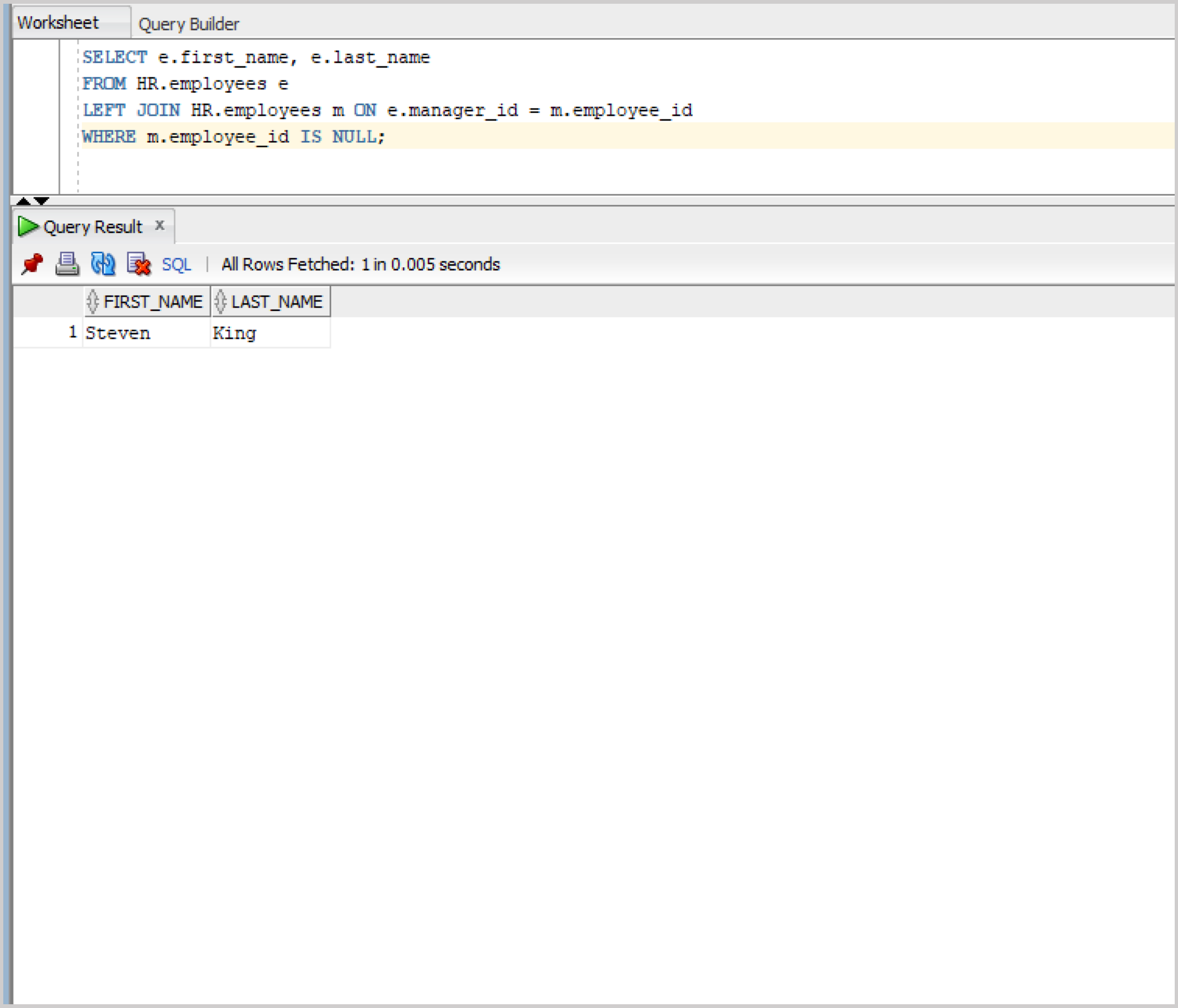
SELECT e.first_name, e.last_name

FROM HR.employees e

LEFT JOIN HR.employees m ON e.manager_id = m.employee_id

WHERE m.employee_id IS NULL;

Snapshot:



Q12)

Query:

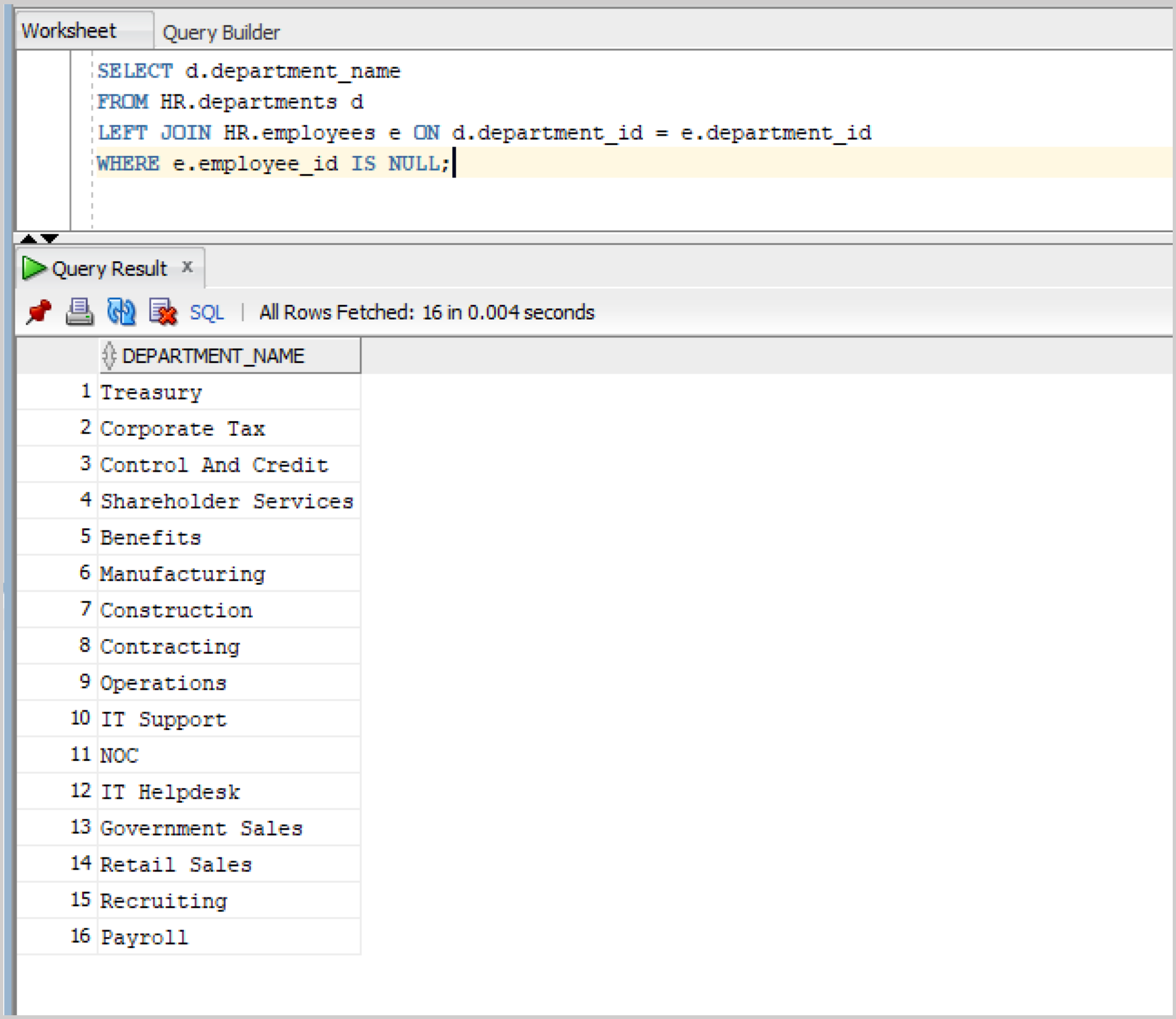
SELECT d.department_name

FROM HR.departments d

LEFT JOIN HR.employees e ON d.department_id = e.department_id

WHERE e.employee_id IS NULL;

Snapshot:



Q13)

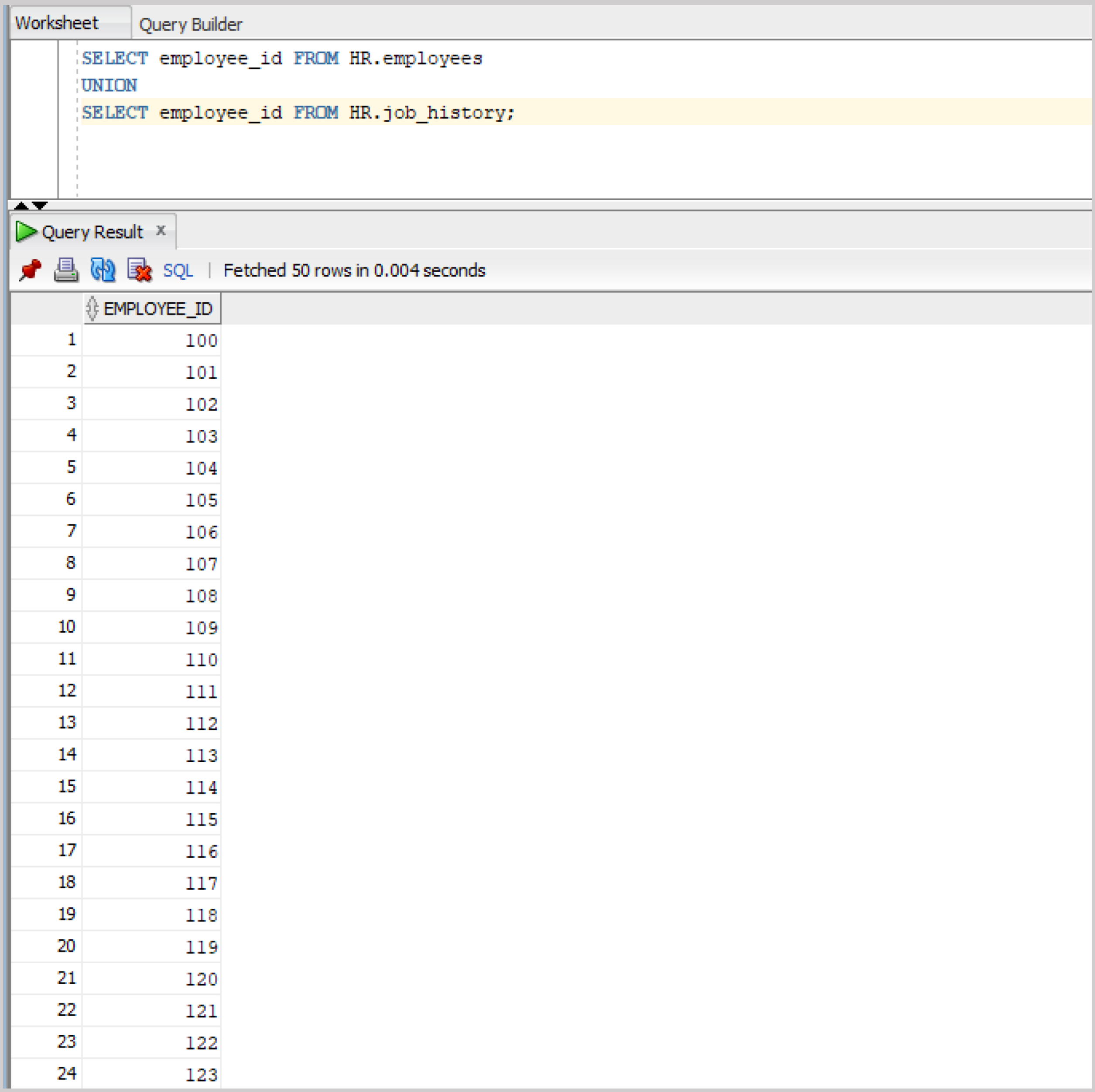
Query:

SELECT employee_id FROM HR.employees

UNION

SELECT employee_id FROM HR.job_history;

Snapshot:



Q14)

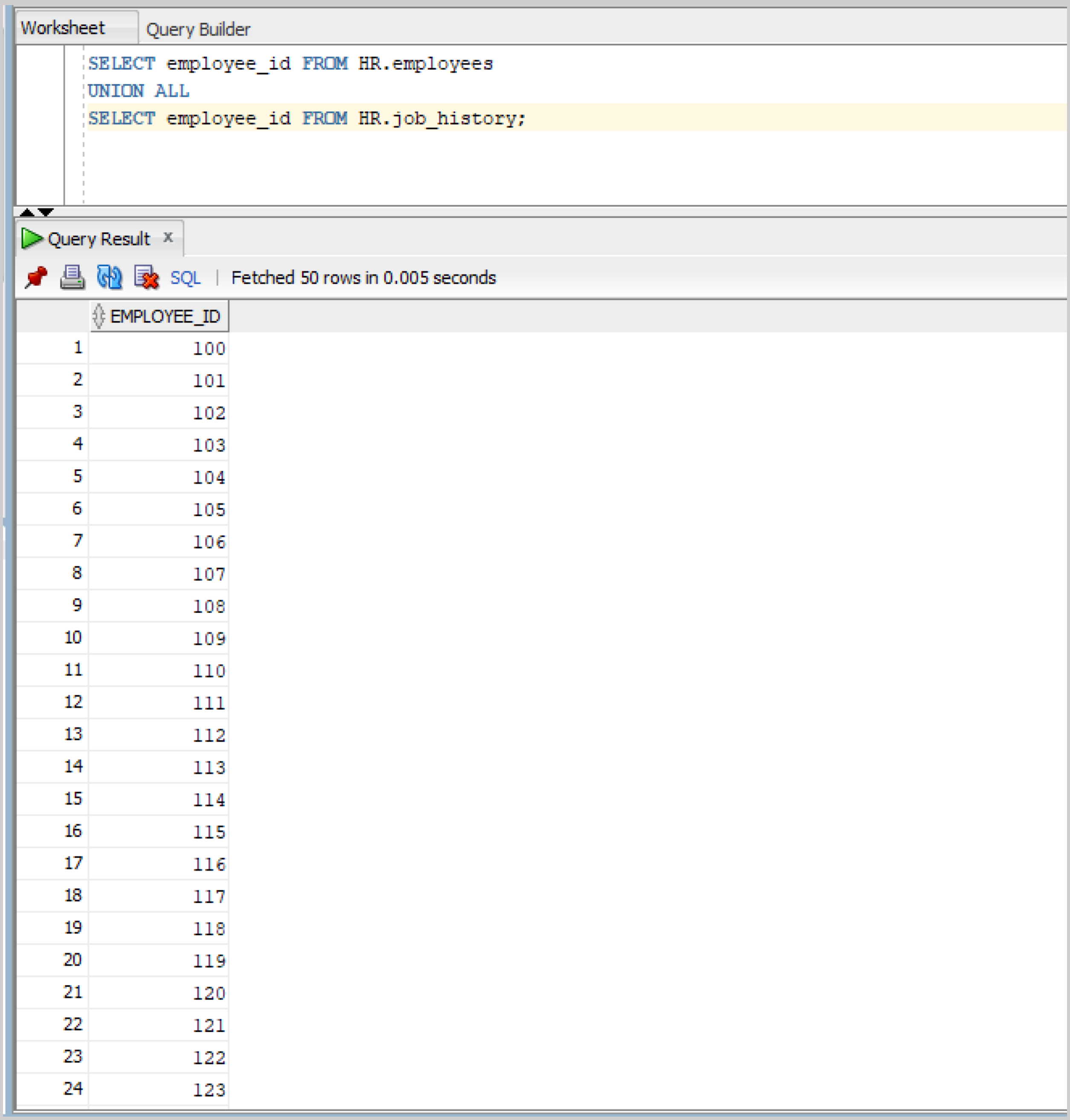
Query:

SELECT employee_id FROM HR.employees

UNION ALL

SELECT employee_id FROM HR.job_history;

Snapshot:



Q15)

Query:

```
SELECT employee_id FROM HR.employees
UNION
SELECT employee_id FROM HR.job_history
ORDER BY employee_id;
```

Snapshot:

Worksheet Query Builder

```sql
SELECT employee_id FROM HR.employees
UNION
SELECT employee_id FROM HR.job_history
ORDER BY employee_id;
```

Query Result

SQL | Fetched 50 rows in 0.008 seconds

	EMPLOYEE_ID
1	100
2	101
3	102
4	103
5	104
6	105
7	106
8	107
9	108
10	109
11	110
12	111
13	112
14	113
15	114
16	115
17	116
18	117
19	118
20	119
21	120
22	121
23	122
24	123

Q16)

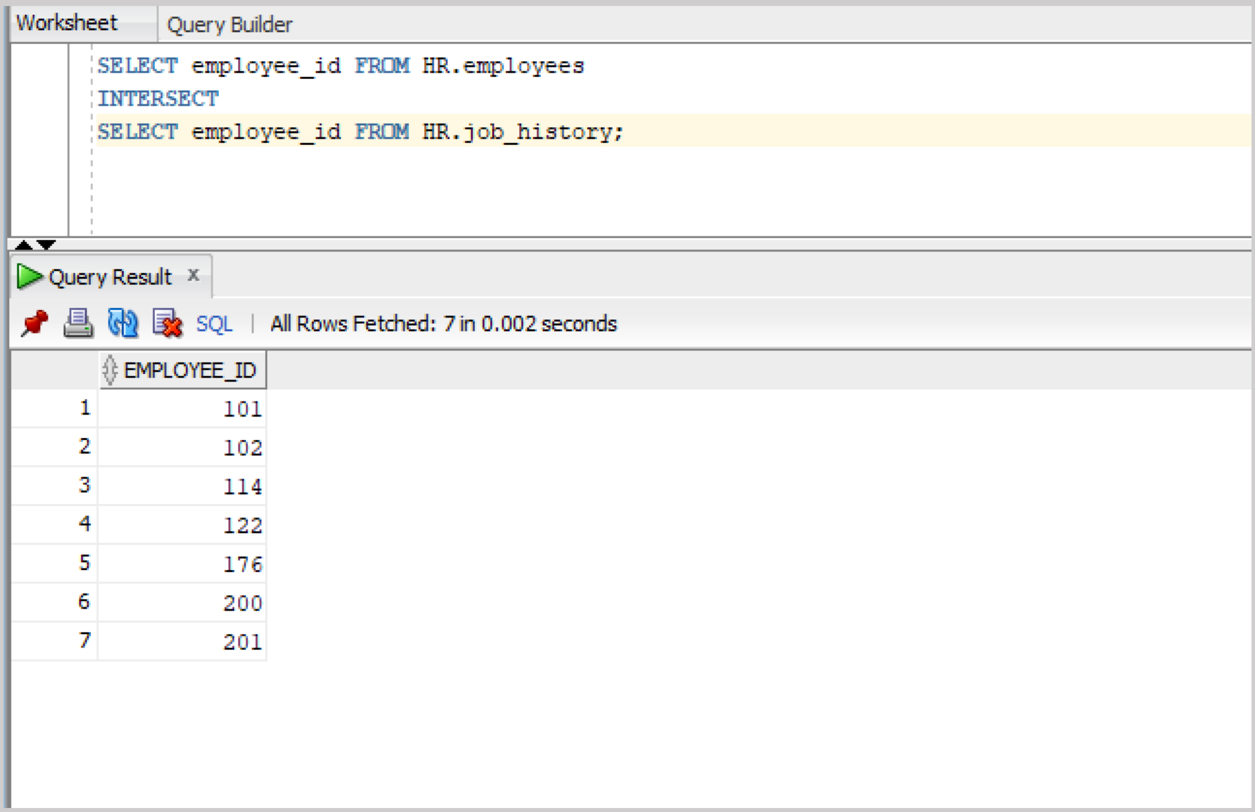
Query:

SELECT employee_id FROM HR.employees

INTERSECT

SELECT employee_id FROM HR.job_history;

Snapshot:



Q17)

Query:

```
SELECT employee_id FROM HR.employees
MINUS
SELECT employee_id FROM HR.job_history;
```

Snapshot:

Worksheet Query Builder

```
SELECT employee_id FROM HR.employees
MINUS
SELECT employee_id FROM HR.job_history;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

	EMPLOYEE_ID
1	100
2	103
3	104
4	105
5	106
6	107
7	108
8	109
9	110
10	111
11	112
12	113
13	115
14	116
15	117
16	118
17	119
18	120
19	121
20	123
21	124
22	125
23	126
24	127

Q18)

Query:

SELECT employee_id FROM HR.employees

UNION

SELECT employee_id FROM HR.job_history;

Snapshot: